

**Sardar Patel University – Student Startup and Innovation Policy –
Navadhārā**

Application Form for Proof of Concept (PoC)/ Prototype Support

Primary Information			
Name of Applicant	:	Prince N. Tagadiya	
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Name of institute with address	:	G H Patel College of Engineering & Technology (GCET), Vallabh Vidyanagar, Anand, Gujarat	
Programme enrolled in	:	B.Tech – ICT Engineering	
Semester	:	Semester IV	
Enrolment No.	:	12402230501034	
Total team size	:	4	
Team Member Information			
1.	Name of Team member	:	Khushi S. Belani
	Date of Birth	:	02/01/2006
	Gender	:	Female
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	Name of institute with address :	G H Patel College of Engineering & Technology (GCET), Vallabh Vidyanagar, Anand, Gujarat
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2.	Name of Team member :	Nisarg G. Patel
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3.	Name of Team member :	Rudra S. Chauhan
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	Programme enrolled in :	B.Tech -ICT Engineering
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Idea/ Prototype Details		
Type of Proposal (PoC/ Prototyping):		PoC
Title of Proposal :		Smart Privacy Health Kiosk: An IoT-Enabled Automated System for Anonymous Access to Essential Healthcare Products
Brief description of Innovation (500 words): Access to essential healthcare products is often limited not by availability, but by embarrassment, social stigma, lack of privacy, and unavailability during emergencies. Many women, students, young adults, travelers, and hostel residents hesitate to purchase products such as sanitary pads, pregnancy test kits, condoms, and other intimate healthcare items due to fear of judgment or discomfort associated with interacting with pharmacy staff. In addition, these products may not always be accessible during late-night emergencies or in locations where pharmacies are unavailable. To address these challenges, we propose the Smart Privacy Health Kiosk, an IoT-enabled, privacy-first automated healthcare access platform designed specifically for Indian conditions. Unlike conventional vending machines, the proposed system focuses on preserving user dignity while improving access to sensitive healthcare products. The kiosk provides 24×7 access to a curated range of healthcare and wellness products including sanitary pads, tampons, menstrual cups, heating patches, pregnancy test kits, ovulation test strips, condoms, lubricants, wet wipes, tissues, first-aid essentials, sanitizers, and other room-temperature healthcare products commonly used in India. The system incorporates a user-friendly 15-inch touchscreen interface with multilingual support and UPI-based cashless payments. Users can browse products, complete secure transactions, and receive their purchases without any human interaction. A key innovation of the proposed solution is the Automatic Privacy Packaging System (APPS). After dispensing, the selected product is automatically placed inside a privacy bag within the machine and transferred to a secure collection locker. This ensures that users can collect sensitive products discreetly, eliminating the fear of social judgment and maintaining complete privacy throughout the purchasing process.		

The kiosk further integrates IoT-enabled smart features including real-time stock monitoring, low-stock notifications, predictive restocking, product jam detection, remote dashboard monitoring, RFID-based maintenance access, tamper detection, and preventive maintenance alerts. Optional fingerprint membership enables faster repeat purchases and controlled access to limited-use benefits such as Emergency Pad Mode.

The system is designed with scalability in mind and supports deployment across colleges, universities, girls' hostels, hospitals, railway stations, bus terminals, airports, shopping malls, corporate offices, and public spaces. Beyond healthcare accessibility, the kiosk also enables sustainable business operations through product sales, digital advertisements, white-label deployments, SaaS subscriptions, franchise opportunities, and annual maintenance services.

The long-term vision is to establish India's first nationwide network of Smart Privacy Health Kiosks that ensures no individual delays addressing an essential healthcare need due to embarrassment, stigma, lack of privacy, or limited access. By combining healthcare accessibility with intelligent technology and human-centered design, the proposed innovation aims to transform how sensitive healthcare products are delivered while preserving dignity, convenience, and inclusivity for all users.

Current Status of Innovation (work done so far) IF ANY:

- Problem identification and market research completed.
- Product categories and deployment locations finalized based on Indian healthcare needs.
- System architecture and feature set defined.
- Revenue model and business feasibility analysis completed.
- Full CAD design of the Smart Privacy Health Kiosk completed prior to pitching.
- Mechanical layout, dispensing mechanism planning, and internal component placement finalized.
- Hardware component selection and preliminary Bill of Materials (BOM) prepared.
- Prototype development and subsystem integration planned as the next phase.

Novelty in your idea:

1. Privacy-First Healthcare Access

Users can purchase sensitive healthcare products without interacting with a shopkeeper, reducing embarrassment and social stigma.

2. Automatic Privacy Packaging System (APPS)

The machine automatically places the selected product inside a **privacy bag** before delivery, ensuring complete discretion.

3. Secure Collection Locker

Purchased products are delivered through a **secure locker system**, so others cannot see what product was bought.

4. Smart & Cashless Experience

The kiosk supports **UPI payments**, a **15-inch multilingual touchscreen**, and a simple user-friendly interface for easy operation.

5. Emergency Pad Mode

In emergency situations, users can receive an **emergency sanitary pad even if payment cannot be completed due to bank server issues**. The pending payment can be completed later, ensuring that immediate healthcare needs are not delayed.

6. Fingerprint-Based Membership

Fingerprint authentication enables **faster repeat purchases**, improves user convenience, and helps prevent misuse of emergency benefits.

7. Automatic Stock Monitoring

The system continuously monitors product availability and automatically sends **low-stock alerts** to operators.

8. Product Jam Detection

If a dispensing failure occurs, the machine can **detect the jam, automatically re-dispense the product, or initiate a refund process**.

9. RFID-Based Maintenance Access

Authorized staff can use **RFID cards** to securely open the machine for refilling and servicing without relying on physical keys.

10. Remote Monitoring & IoT Connectivity

Operators can remotely monitor **inventory levels, sales reports, machine health, and maintenance alerts** from anywhere through an IoT-enabled dashboard.

11. Predictive Restocking

The system analyzes usage patterns to **forecast future demand and optimize refill schedules**, reducing stock-out situations.

12. Healthcare Awareness Portal

Users can access **QR-based educational content** related to menstrual hygiene, pregnancy guidance, safe-sex awareness, and women's health.

13. Interactive Ambient Lighting

The internal display and dispensing area automatically illuminate when users interact with the touchscreen, improving visibility and creating a premium user experience.

14. Advertisement & White-Label Support

The kiosk supports **digital advertisements** and can be customized for hospitals, colleges, NGOs, and corporate organizations through white-label deployment.

15. Multi-Kiosk Management

A centralized system allows operators to manage **multiple kiosks simultaneously**, including inventory synchronization and consolidated reporting.

The uniqueness of this innovation lies in combining privacy, emergency healthcare support, smart technology, and sustainable business operations into a single healthcare access platform designed specifically for Indian users.

Will your idea/ solution impact the population and economy?

Yes.

The proposed Smart Privacy Health Kiosk has the potential to create both **social and economic impact**.

Impact on Population

- Improves access to essential healthcare and hygiene products in a private and dignified manner.
- Reduces embarrassment and social stigma associated with purchasing sensitive healthcare products.
- Supports women during menstrual emergencies by providing 24×7 access to sanitary products.
- Increases awareness of menstrual hygiene, reproductive health, and safe healthcare practices.

- Benefits students, women, travelers, hospital visitors, and people in public spaces who require immediate access to healthcare products.
- Ensures that healthcare needs are not delayed due to lack of availability or privacy concerns.

Impact on Economy

- Creates opportunities for manufacturing, installation, servicing, and maintenance of kiosks.
- Generates employment through operations, refilling, and support services.
- Encourages the growth of health-tech and smart vending businesses in India.
- Provides additional revenue through product sales, digital advertisements, white-label deployments, and franchise models.
- Supports the vision of developing scalable, indigenous healthcare solutions under the "Make in India" initiative.

Overall, the proposed solution aims to improve public health outcomes while creating a sustainable and economically viable business model with the potential for nationwide deployment.

Do you envisage your Idea/ Innovation to be a Startup?:

Yes.

The proposed **Smart Privacy Health Kiosk** has strong potential to evolve into a sustainable startup. The solution addresses a real problem faced by millions of people who require private and immediate access to essential healthcare products.

The startup can generate revenue through:

- **Product sales** from healthcare and hygiene products.
- **Digital advertisements** displayed on the kiosk screen.
- **White-label deployments** for hospitals, colleges, NGOs, and corporate organizations.
- **Franchise opportunities** for large-scale expansion.
- **Annual Maintenance Contracts (AMC)** and servicing.
- **Dashboard subscription services** for institutions managing multiple kiosks.

The long-term vision is to establish a **nationwide network of Smart Privacy Health Kiosks**, improving healthcare accessibility while building a scalable and financially sustainable health-tech business in India.

Which of the following facilities do you wish to avail from SPU-SSIP Navadhārā?

Choose as many as you like;

A: Co-working Space ✓

B: Mentorship and Guidance (Faculty Mentor: Dr. Geetali Saha, EC Department, GCET)

C: Networking Opportunities ✓

D: Access to Funding and Grants ✓

E: Patent and IPR Support ✓

F: Other infrastructure support at SPU ✓

Please specify: Electronics prototyping lab, testing equipment

Fund Requirement Details

Breakup of Financial Support Requested with details of required instruments/ gadgets/ etc:

Sr. No.	Item	Purpose	Amount (₹)
1	Industrial Prototype Cabinet & Fabrication	Full-scale kiosk enclosure and structural assembly	35,000
2	Raspberry Pi 5 (8GB), SSD & Accessories	Main controller, touchscreen operation and system integration	12,000
3	15-inch Touchscreen Display	Product browsing, multilingual interface and advertisements	10,000
4	ESP32 Controller Boards	Real-time control of sensors and dispensing units	2,000
5	Dispensing Mechanism (Motors, Drivers, Springs & Mechanical Components)	Multi-product dispensing system	20,000
6	Automatic Privacy Packaging System (APPS) Components	Privacy bag dispensing and packaging mechanism	15,000
7	Secure Collection Locker & Electronic Lock	Discreet product collection	8,000
8	Sensors (IR, Tamper, Door, Locker, Tilt & Vibration)	Safety, monitoring and fault detection	8,000
9	RFID Access System & Fingerprint Scanner	Secure maintenance access and membership features	6,000
10	4G/Wi-Fi Communication Module	IoT connectivity and remote monitoring	4,000
11	Industrial Power Supply & Protection Circuit	Reliable power distribution and protection	8,000
12	Battery Backup System	Continued operation during power failures	18,000
13	Solar Panel & Solar Charge Controller	Renewable power support and rural deployment readiness	15,000
14	Electronics Integration Components (PCB, Wiring, Connectors, Mounts)	Hardware interfacing and assembly	7,000
15	Prototype Testing, Validation & Pilot Deployment	Functional testing and real-world evaluation	10,000
16	Research & Development (R&D) Expenses	Design optimization, iterations and performance improvements	20,000
17	Contingency & Miscellaneous Expenses	Unforeseen requirements during development	10,000

Total Amount of Financial Support Requested:

₹2,30,000 (Rupees Two Lakh Thirty Thousand Only)

The requested financial support is intended for the development of an **industrial-grade Proof of Concept (PoC) prototype** of the Smart Privacy Health Kiosk rather than a basic academic demonstration model.

The proposed prototype will integrate privacy-focused dispensing, automated packaging, secure collection, IoT-enabled monitoring, remote management, and emergency healthcare accessibility into a single deployable system. The funding will support hardware procurement, fabrication, integration, testing, validation, and iterative R&D required to evaluate the technical and commercial feasibility of the solution.

The developed prototype will serve as a foundation for pilot deployments in institutions such as colleges, hospitals, hostels, and public spaces, enabling future cost optimization and large-scale commercialization.

The objective is not merely to build a machine, but to validate a scalable healthcare access platform with the potential to evolve into a sustainable Indian health-tech startup capable of improving healthcare accessibility while preserving dignity and privacy.

Required Documents;

- **Bonafide Certificate from College/ Department.**
- **Passport size photograph of Team members.**
- **Undertaking (in prescribed format, once the Project is approved by SSIP-SPU-Navadhārā).**

Any other information/ detail (If any):

Our team has prior experience in developing and executing innovation projects with successful outcomes.

- We have previously worked on an **SSIP-funded agricultural drone project** focused on supporting farmers through technology-driven solutions.
- We secured the **Runner-Up position at the India Level BGI Hackathon 2026 (Vision 2047: Viksit Bharat)** among **900+ participants from across India** for our **IoT-enabled Water Leakage & Theft Detection System**.
- We were also **winners at the State-Level TECHNOVATION 2026 (Science-to-Systems Hackathon)** for developing an innovative safety solution under the theme of Disaster Management and Safety.
- Our earlier work on a **Density-Based Traffic Light Controller** was recognized through publication in **Electronics For You (EFY)** magazine.
- Through these experiences, our team has gained practical exposure in **IoT system development, prototype fabrication, sensor integration, real-time monitoring, dashboard development, field validation, and transforming ideas into working solutions**.

These achievements demonstrate our capability to successfully execute the proposed Smart Privacy Health Kiosk project and translate it from a prototype into a scalable and impactful solution.

Applicant signature with Name

Date: